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Project PHEX '27

Machine Learning & Edge Computing enabled Energy Management System for optimization of Hybrid Electric Powertrain: Proof-of-Concept and Predictive EMS

by

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Performed at

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Date Submitted: 13/06/2026

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DECLARATION

I, Mishael Julian, declare that this internship report is an original record of the work carried out during the Project PHEX '27 internship period and is submitted in partial fulfilment of the Bachelor of Technology requirements.

The work described in this report is based on the internship Proof-of-Concept scope, approved technical references, simulation outputs, communication studies, benchmarking activities, and documentation tasks assigned during the internship period.

Signature: _____

Date: 13/06/2026

ACKNOWLEDGEMENT

Firstly, I would like to thank Mr. Martin Dsouza, the Student Project Director and Technical Head of Project PHEX '27, for giving me the opportunity to undergo the internship program and for continuously guiding me, mentoring me on a technical level, encouraging, and supporting me throughout the project. Insights regarding Hybrid Electric Vehicle Energy Management Systems and the project development process proved to be very helpful in completing the project successfully.

I am grateful to the faculty coordinators, Dr. Pradeep GS and Dr. Parag Jose, for guiding and encouraging me throughout the course of the internship period.

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I would also like to extend my thanks to the School of Engineering and Technology, CHRIST (Deemed to be University), Bengaluru for offering me the academic setting and all required resources that made possible to perform this internship on a technical engineering project.

Lastly, I would like to say thanks to everybody who, whether directly or indirectly, contributed to the successful completion of this internship and supported my studies in such domains as Artificial Intelligence, Machine Learning, Hybrid Electric Vehicles, and Energy Management Systems.

ABSTRACT

This project was conducted within the context of Project PHEX '27, which sought to develop a PoC Energy Management System (EMS) for a PHEV. The major aim here is the creation of an EMS system that will optimize the use of energy between the ICE and the electric propulsion system, depending on various driving situations.

Some of the activities include modeling of the vehicle, battery, motor, engine, and generator, alongside developing a rule-based EMS, which entails managing the state-of-charge, regenerative braking system, engine efficiency, torque-splitting, and prediction concepts. Machine learning has also been used in developing predictive modules for the EMS, where traffic and route contexts are considered along with future energy consumption predictions.

Simulation studies were conducted using representative drive cycles, including WLTP Urban, WLTP Mixed, and Bangalore Urban Driving Cycles. System performance was evaluated using key performance indicators such as fuel consumption, state-of-charge deviation, engine runtime, energy recuperation, and operating efficiency. The developed framework successfully demonstrated the feasibility of integrating rule-based supervisory control with predictive EMS concepts for hybrid vehicle energy optimization.

The internship provided practical exposure to hybrid vehicle architectures, energy management strategies, simulation-driven development, machine learning assisted decision support systems, and automotive control methodologies. The resulting Proof-of-Concept establishes a technical foundation for future expansion and integration within the broader objectives of Project PHEX '27.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
EMS	Energy Management System
HEV	Hybrid Electric Vehicle
PHEV	Plug-in Hybrid Electric Vehicle
SOC	State of Charge
DAQ	Data Acquisition
CAN	Controller Area Network
CAN FD	Controller Area Network Flexible Data-rate
ML	Machine Learning
KPI	Key Performance Indicator
PoC	Proof-of-Concept
DCT	Dual-Clutch Transmission

2. PROJECT BACKGROUND AND REFERENCE VEHICLE

2.1 Internship Context

The project "PHEX '27" entails continuous efforts in researching and developing solutions in regard to efficient energy management in hybrid electric and plug-in hybrid electric vehicles. Within this project, the internship was conducted to create a Proof-of-Concept (PoC) energy management system that could optimize the cooperation between internal combustion engine, electric motor, the vehicle's battery system, and other vehicle sub-systems.

This internship concentrated on the creation of the energy management system (EMS) architecture within a power-split hybrid system that can be used in a vehicle similar to the Mercedes-Benz A250e prototype vehicle. The developed system uses rule-based control systems along with predictive decision-making concepts to maximize the energy consumption in various traffic situations. Among main features of the developed EMS there are SOC (State-of-Charge) management, start-stop engine control, torque splitting management, regeneration control, energy saving in traffic, battery saving by route planning, as well as mode switching by efficiency.

Multiple simulation modules were created for analysis of the suggested EMS structure, consisting of vehicle dynamics simulation, battery model, motor model, engine model, generator model, and drive cycle simulation. Predictive functions based on machine learning were included to predict future vehicle demands and traffic conditions, allowing the EMS to make context-dependent decisions related to the [energy management](#). The test was conducted by means of several drive cycles such as WLTP Urban drive cycle, WLTP Mixed drive cycle, and Bangalore Urban Driving Cycle; several criteria were considered during analysis, including fuel economy, SOC variation, energy regeneration rate, efficiency, and engine on time.

The results obtained during this internship are a base for further development of Project PHEX '27 phases when it comes to more sophisticated predictive technologies, vehicle data acquisition systems, vehicle communication architecture, and energy optimization for vehicles.

2.2 Reference Vehicle and Technical Baseline

Parameter	Reference Value	Notes / Source
Reference Vehicle	Mercedes-Benz A250e PHEV Sedan	Primary benchmark reference
Vehicle Class	Compact Passenger Car / Autonomous Passenger Vehicle	Project target class
Vehicle Mass	1800 kg	Kerb mass plus occupants/luggage as required
Battery Capacity	15 kWh	Reference baseline
Initial SOC	70%	PoC assumption
Minimum SOC	20%	Operating lower limit
Electric Motor Power	80 kW	Reference value
ICE Power	120 kW	Reference value
Combined Max Power	160–180 kW	Reference range
Drivetrain	Power-split PHEV	Architecture baseline
Wheel Radius	0.31 m	Reference assumption
Drag Coefficient (Cd)	0.22	Aerodynamic parameter
Frontal Area	2.2 m ²	Aerodynamic parameter
Rolling Resistance	0.012	Road-load assumption
Top Speed	210 km/h	Reference assumption
EV-only Threshold	<45% torque demand	Control logic assumption
Engine Activation SOC	<25% SOC	Control logic assumption

2.3 Project Objective

One of the major goals of this internship has been the design and validation of the Proof-of-Concept (PoC) Energy Management System (EMS) of the power-split Plug-in Hybrid Electric Vehicle (PHEV) following the Mercedes-Benz A250e architecture. This involved devising the EMS to be able to monitor the energy exchange among the internal combustion engine, electric motor, generator, and battery unit in order to minimize the fuel consumption while maximizing the system's efficiency and keeping the battery State-of-Charge (SOC) at safe levels. Apart from employing rule-based EMS concepts such as regenerative braking, torque split control, engine start-stop, and SOC management techniques, the project also examined the application of predictive approaches to EMS, incorporating knowledge about the road characteristics and traffic situation, as well as applying demand prediction based on machine learning algorithms. The effectiveness of the designed and implemented EMS was tested via simulation using WLTP Urban, WLTP Mixed, and Bangalore Urban driving cycles, evaluating its performance in terms of fuel efficiency, SOC deviation, duration of engine operation, and energy recovery. The developed proof-of-concept model can be used as the basis for future integration of advanced systems in Project PHEX '27.

3. REPORT STRUCTURE

The following chapter flow should be used in the final internship report. Each intern should write only the sections relevant to their contribution, while maintaining the same overall report architecture.

Chapter	Expected content
Chapter 1	Introduction and Project Objective
Chapter 2	Project Background and Reference Vehicle
Chapter 3	HEV EMS Proof-of-Concept Architecture
Chapter 4	Intern-Specific Methodology and Development
Chapter 5	Simulation / Communication / Benchmarking Methodology
Chapter 6	Results and Validation
Chapter 7	Discussion, Challenges, and Learning Outcomes
Chapter 8	Technical Outputs and Deliverables
Chapter 9	Conclusion and Future Scope / Project PHEX Integration



3 HEV EMS PROOF-OF-CONCEPT ARCHITECTURE

3.1 Overview of the HEV EMS Proof-of-Concept

HEV's use several different energy sources, usually an ICE engine along with a motor that drives the wheels electrically. Coordination of these multiple energy sources efficiently is done by using an EMS. The main function of the EMS is to increase fuel efficiency and at the same time decrease emissions, while taking care of battery state of charge and providing satisfactory vehicle performance [1].

As a part of the Project PHEX '27, an intelligent PoC HEV's energy management system was developed using both rule-based and predictive decision-making techniques. The design of the developed system uses a combination of features such as traffic-aware route analysis, predicting power demands, battery SOC management, supervisory mode selection, safety validation, and torque distribution.

In contrast with the conventional reactive approach to emergency mobility systems, the architecture introduced will have prediction layers based on knowledge of routes and traffic to help predict the future operating environment. In this way, the EMS will be able to make forward-looking decisions on how to use battery power, engine efficiency, and energy harvesting.



Figure 3.1: Overall control hierarchy of the Project PHEX Hybrid Energy Management System.

3.2 Overall System Architecture

In the proposed EMS architecture, the EMS system is designed with a hierarchical control scheme, where the upper layers' planning models provide strategies that are successively improved into specific powertrain instructions. The information flows from prediction and planning levels to the actual powertrain controllers.

At the lower layer, the Traffic and Route Prediction model analyzes the information about roads and traffic conditions. After that, the information is passed to the Demand Forecasting model that forecasts vehicle power demand based on driving conditions prediction.

As mentioned above, the architecture of the EMS incorporates a hierarchical control design where the decisions made at higher levels are transformed into practical commands executed by lower powertrain control modules. The flow of information proceeds from prediction and planning modules to the physical powertrain control modules successively.

The first of the mentioned modules, Traffic and Route Prediction, takes into account route data and predicts possible traffic states. The results are then used by the Demand Forecasting module to predict the future power needs of vehicles.

The Strategic SOC Planning module utilizes route and demand data to compute the desired battery usage levels. This is analyzed by the Decision Engine and is used along with predictive recommendations to decide the appropriate operating mode.

All the decisions undergo analysis within a Safety Override Layer before execution to ensure that system constraints are met and safe operating conditions maintained. The appropriate operating mode is sent to the Torque Split Controller, which assigns engine and electric motor power demands. Subsequently, the powertrain systems execute the decision using the engine, electric motor, battery, and generator subsystems.

The hierarchical structure provides a clear division between strategic planning and actual execution processes, maintaining modularity and scalability of the system [2].

3.3 Detailed Data Flow Architecture

The EMS system is comprised of several interlinked software blocks that exchange information about the vehicle state, predictions, and control signals. This can be seen in Figure 3.2.

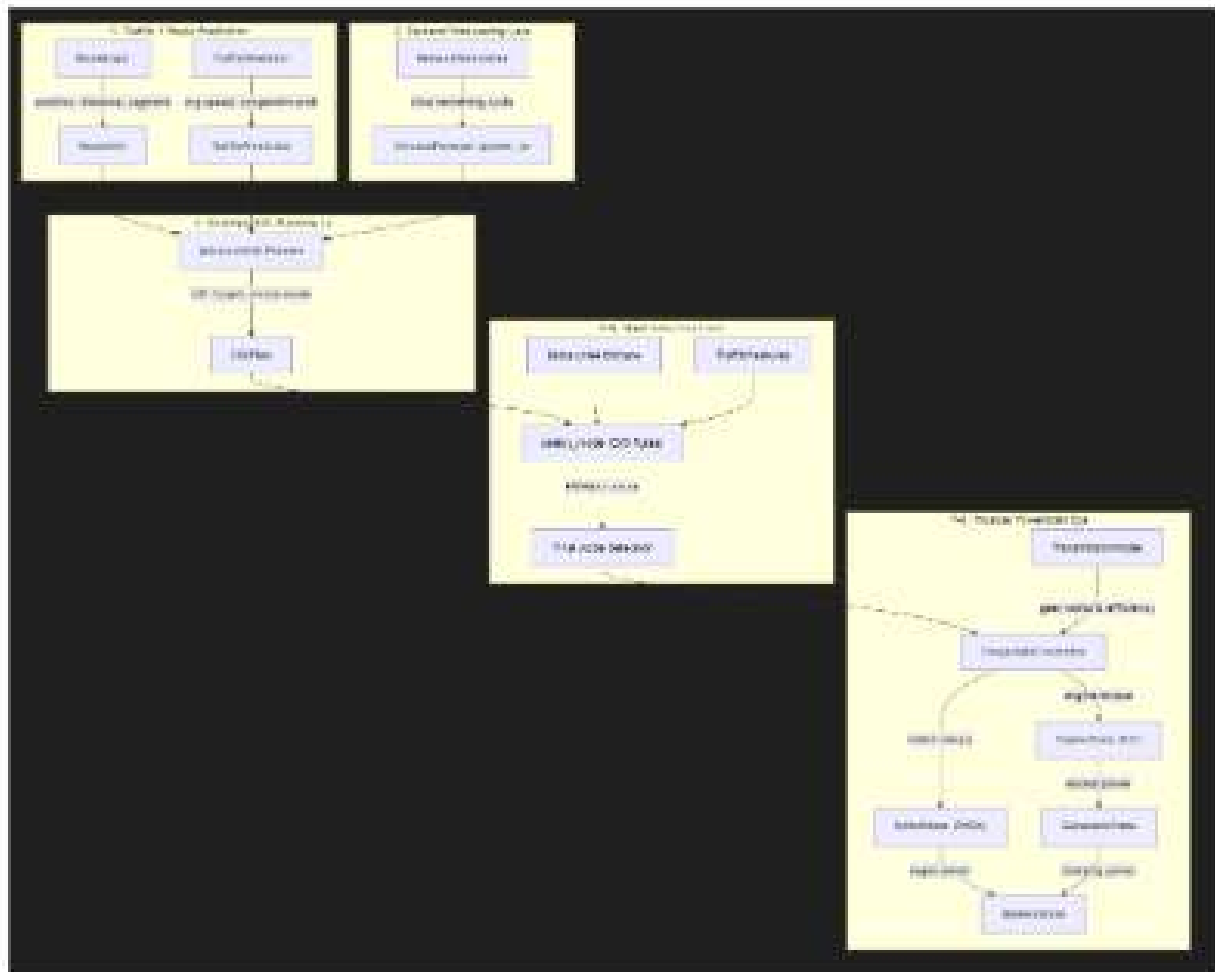


Figure 3.2: Detailed information flow among the prediction, planning, decision, and execution layers.

The Route Logic block supplies position and route segment details to the Traffic Predictor. The traffic feature, including predictions about congestion levels and traffic conditions, is formed and sent to the Demand Forecasting layer.

The Demand Forecasting module calculates future power demand features such as load trends and confidence measures. This forecast data is given to the SOC Planner that computes battery charge retention goals and future operation guidelines.

The Decision Engine gets battery condition data, traffic conditions, and SOC Planning results. Then, based on this information, the supervisory controller chooses the right Energy Management Mode. Then this mode will be further considered by the Torque Splitter module that sets the share of each power train component.

At the hardware implementation level, the coordination between engine torque, motor torque, energy recovery, and charging is performed to meet the demand of the vehicle under constraints.

The use of modularity for a data flow architecture facilitates further improvements to the system since modifications may be performed independently of other parts of the system.

3.4 Software Architecture

The software implementation of the EMS PoC follows a modular architecture developed in Python. Individual components were organized into separate modules to improve maintainability, scalability, and testing capability.

Figure 3.3: Software module organization of the Project PHEX EMS implementation.

The major software components are summarized below:

Module	Primary Reason
route_logic.py	Route segmentation and route-context processing
traffic_predictor.py	Traffic condition estimation and congestion analysis
demand_forecaster.py	Future power demand prediction
soc_planner.py	Strategic battery SOC planning
decision_engine.py	EMS supervisory mode selection
safety_override.py	Constraint validation and protection logic
torque_split.py	Power distribution between ICE and electric motor
engine_model.py	Internal combustion engine representation
motor_model.py	Electric traction motor representation
battery_model.py	Battery pack behaviour and SOC tracking
generator_model.py	Generator charging subsystem
transmission_model.py	Torque transfer and drivetrain behaviour
simulator.py	Drive-cycle execution environment
evaluation.py	KPI computation and performance assessment

The modular design enabled independent testing of subsystems and facilitated integration of predictive intelligence components alongside conventional rule-based control strategies. Furthermore, the architecture supports future migration toward embedded automotive controllers and real-time vehicle applications.

3.5 Chapter Summary

In this chapter, the architecture of the proof-of-concept of Project PHEX Hybrid Energy Management System was explained. An EMS architectural structure that consisted of seven layers, namely route prediction, traffic analysis, demand forecasting, state of charge planning, supervisory control, safety verification, and torque allocation, was established. The EMS architecture serves as a guide for creating an intelligent hybrid vehicle management system. The following chapters detail the process involved in the creation and evaluation of the EMS.-

4. INTERN-SPECIFIC METHODOLOGY AND DEVELOPMENT

4.1 HEV Simulink Architecture

The Hybrid Electric Vehicle Energy Management System (HEV EMS) Proof-of-Concept was implemented as a modular software simulation environment developed in Python. The architecture was designed to emulate the major subsystems of a power-split Plug-in Hybrid Electric Vehicle (PHEV) while providing a flexible platform for testing energy management strategies.

The simulation framework consists of multiple interconnected subsystem models representing the vehicle, engine, electric motor, battery pack, generator, transmission system, and supervisory energy management controller. Each subsystem operates independently while exchanging information through a centralized vehicle state representation.

The modular design allows individual components to be modified, tested, and validated without affecting the remaining architecture. This approach mirrors the subsystem-oriented design methodology commonly adopted in automotive control system development [1].

Vehicle Longitudinal Model

The vehicle model represents the physical motion of the reference PHEV platform. It computes vehicle speed, acceleration, traction demand, aerodynamic drag, rolling resistance, and road-load effects. The model serves as the primary interface between the drive cycle and the powertrain simulation.

Engine Model

The Internal Combustion Engine (ICE) model simulates fuel-powered propulsion. The model calculates engine power output, operating state, and efficiency characteristics based on the power request generated by the EMS controller.

Electric Motor Model

The electric traction motor model provides propulsion assistance during EV and Hybrid operating modes. Motor torque requests are received from the torque split controller and converted into electrical energy consumption.

Battery Model

The battery subsystem maintains State-of-Charge (SOC) estimation throughout simulation execution. Charging and discharging behaviour are tracked continuously, allowing the EMS controller to make battery-aware operating decisions.

Driver Demand Model

Driver demand is derived from standardized drive cycles. Vehicle speed requirements are converted into wheel power demand and subsequently propagated through the drivetrain model.

Signal Routing Between Subsystems

Information exchange between subsystem models is performed through structured vehicle state variables. This includes SOC measurements, vehicle speed, traffic features, route context, power demand, operating mode selection, and torque requests.

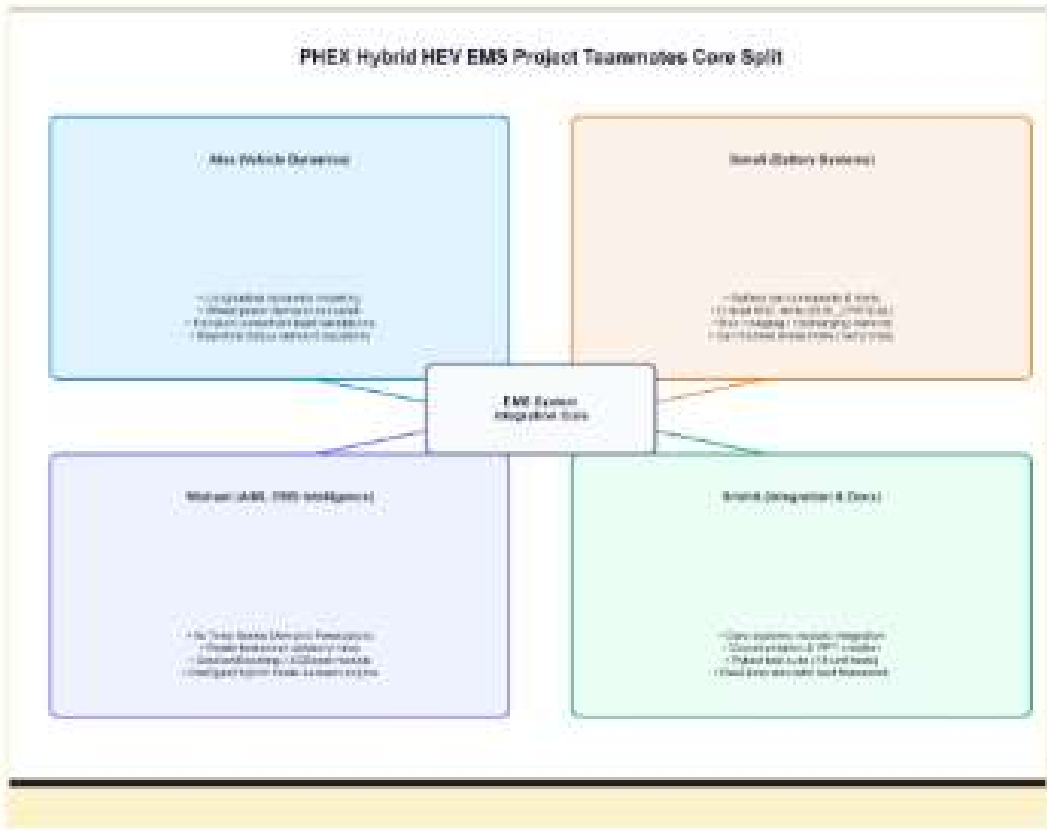


Figure 4.1: HEV EMS subsystem interaction architecture.

4.2 Rule-Based EMS Logic

The supervisory controller utilizes a rule-based Energy Management Strategy (EMS) derived from Project PHEX requirements. The objective of the EMS is to determine the optimal operating mode while balancing fuel consumption, battery utilization, drivability, and energy recovery.

The EMS controller evaluates the complete vehicle state at each simulation step and selects one of several operating modes including EV Mode, Hybrid Assist Mode, Regenerative Braking Mode, Charge Sustain Mode, and ICE Mode.

SOC Management Logic

Battery State-of-Charge serves as a primary control variable within the EMS architecture. Strategic SOC targets are maintained to ensure sufficient battery availability for urban driving conditions while preventing excessive battery depletion.

The controller continuously evaluates battery SOC and adjusts operating modes to preserve battery health and maintain energy reserves.

Engine ON/OFF Logic

The engine activation strategy is based on battery condition, vehicle operating state, route context, and power demand requirements. The controller enables engine operation when electric propulsion alone is insufficient or when battery charging is required.

Torque Split Logic

Power demand generated by the vehicle model is distributed between the internal combustion engine and electric motor using a supervisory torque allocation mechanism.

The controller determines:

- Engine torque contribution
- Motor torque contribution
- Generator charging power
- Battery charging/discharging behaviour

This mechanism allows efficient utilization of both propulsion sources while maintaining system operating constraints.

Regenerative Braking Logic

During vehicle deceleration events, the EMS activates regenerative braking mode. Instead of dissipating kinetic energy as heat, the traction motor operates as a generator and converts vehicle kinetic energy into electrical energy stored within the battery pack.

The regenerative controller remains active until battery charging limits or operating constraints are reached.

Implementation of Martin's EMS Supervisory Rules

The final EMS implementation incorporates eight supervisory control rules defined within Project PHEX:

- 1. Urban Stop-and-Go EV Priority
- 2. SOC Preservation Before Urban Zones
- 3. Traffic-Aware Mode Planning
- 4. GPS Route Preview Planning
- 5. ICE Efficiency Sweet Spot Optimization
- 6. Regenerative Braking Priority
- 7. Maximum Acceleration Hybrid Assist
- 8. Low SOC Recovery Charging

Each rule was implemented within the decision engine and verified using dedicated unit and integration tests.

Table 4.1: EMS Supervisory Rule Compliance Matrix

Rule ID	Description	Implemented
R1	Regenerative Braking	Yes
R2	Idle Stop	Yes
R3	Critical SOC Charging	Yes
R4	Thermal Protection	Yes
R5	City EV Mode	Yes
R6	Highway Engine Mode	Yes
R7	Hybrid Assist	Yes

4.3 ML-Assisted Predictive EMS Concept

A predictive EMS layer was developed to extend the capabilities of the traditional rule-based controller.

Unlike conventional EMS systems that react solely to current vehicle conditions, the predictive layer attempts to anticipate future operating requirements through route and traffic analysis.

The predictive framework consists of three primary modules:

Traffic Prediction Layer

The Traffic Predictor estimates future traffic conditions using route segment information and historical traffic characteristics.

Generated features include:

- Expected congestion level
- Average segment speed
- Route classification
- Traffic severity estimation

Demand Forecasting Layer

The Demand Forecaster predicts future vehicle power demand using time-series analysis techniques.

The forecasting subsystem generates:

- Future power estimates
- Peak demand forecasts
- Prediction confidence metrics

These forecasts provide the EMS with additional situational awareness for future operating conditions.

Strategic SOC Planning Layer

The SOC Planner combines traffic forecasts and demand predictions to generate battery utilization recommendations.

The planner determines:

- Desired SOC target
- Charge preservation recommendations
- Future operating strategy suggestions

The predictive layer remains a Proof-of-Concept implementation intended to demonstrate future intelligent EMS capabilities rather than production deployment.

Figure 4.2: Predictive EMS Control Hierarchy.

4.4 Drive Cycle Testing

The developed EMS was evaluated using multiple representative driving cycles to assess controller performance under different operating conditions.

WLTP Urban Cycle

The WLTP Urban cycle represents low-speed city driving characterized by frequent acceleration, deceleration, and stop-and-go operation.

WLTP Mixed Cycle

The WLTP Mixed cycle combines urban and highway driving segments to evaluate adaptive EMS behaviour across varying operating conditions.

Bangalore Driving Cycle (BMDC)

A custom Bangalore Metropolitan Driving Cycle was utilized to approximate real-world Indian traffic conditions. This cycle contains higher congestion levels and increased stop-and-go operation compared with standard drive cycles.

Drive cycle execution was performed using the simulator framework developed within Project PHEX.

4.5 KPI Evaluation

Controller performance was evaluated using a collection of Key Performance Indicators (KPIs). The primary KPIs include:

Fuel Consumption

Total fuel utilized during drive cycle execution.

SOC Deviation

Variation in battery State-of-Charge throughout the simulation.

Engine Runtime

Percentage of simulation time during which the engine remained active.

Energy Recuperation

Electrical energy recovered through regenerative braking operation.

Estimated Operating Cost

Approximate operating cost derived from energy and fuel consumption metrics.

The calculated KPIs provide quantitative measures for comparing EMS operating strategies and evaluating overall system effectiveness.

5. SIMULATION / COMMUNICATION / BENCHMARKING METHODOLOGY

5.1 Simulation Framework

The Project PHEX Hybrid Energy Management System was evaluated using a software-based simulation environment developed in Python. The simulation framework integrates vehicle dynamics, powertrain subsystem models, supervisory control logic, predictive planning modules, and performance evaluation components into a unified testing platform.

The simulator executes drive-cycle data sequentially and updates the vehicle state at each simulation step. Based on the current operating conditions, the Energy Management System selects an operating mode and determines the distribution of power between the internal combustion engine, electric motor, battery system, and generator subsystem.

The primary objective of the simulation framework is to provide a controlled environment for validating EMS behaviour prior to deployment on embedded automotive hardware.

Figure 5.1: Simulation execution workflow.

5.2 Simulation Assumptions and Initial Conditions

To ensure repeatable and comparable results, a common set of simulation assumptions was adopted throughout all experiments.

Vehicle Parameters

The simulation was based on the reference Mercedes-Benz A250e Plug-in Hybrid Electric Vehicle introduced in Chapter 2. Vehicle mass, battery capacity, motor rating, engine rating, aerodynamic characteristics, and drivetrain assumptions were maintained consistently across all test scenarios.

Battery Parameters

The following battery operating assumptions were used:

Parameter	Value
Initial SOC	70%
Minimum SOC	20%
Battery Capacity	15 kWh
EV Priority Threshold	<45% Torque Demand
Engine Recovery Threshold	<25% SOC

Traffic and Route Assumptions

Route segmentation and traffic conditions were generated using predefined simulation scenarios representing urban, mixed, and congested driving environments. These conditions were supplied to the predictive EMS modules during simulation execution.

Environmental Assumptions

The Proof-of-Concept assumes:

- Flat road conditions
- No elevation changes
- Constant ambient temperature
- No battery aging effects
- No component degradation effects

These simplifications were considered acceptable for the objectives of the internship PoC.

5.3 Drive Cycle Test Methodology

The Energy Management System was evaluated using multiple drive cycles representing different vehicle operating environments.

WLTP Urban Drive Cycle

The WLTP Urban cycle was selected to evaluate EMS performance during low-speed city driving conditions characterized by frequent acceleration, braking events, and traffic congestion.

This scenario primarily evaluates:

- EV mode utilization
- Regenerative braking effectiveness
- Urban SOC preservation behaviour

WLTP Mixed Drive Cycle

The WLTP Mixed cycle combines urban and highway driving segments. This cycle evaluates the EMS ability to transition between operating modes while balancing fuel efficiency and battery utilization.

This scenario primarily evaluates:

- Hybrid mode selection
- Highway engine utilization
- Strategic SOC planning

Bangalore Metropolitan Driving Cycle (BMDC)

A custom Bangalore Driving Cycle was developed to approximate real-world Indian urban traffic conditions.

Characteristics include:

- High stop-and-go frequency
- Extended congestion periods
- Frequent acceleration and braking events
- Increased regenerative braking opportunities

The BMDC was used to evaluate the effectiveness of the predictive and traffic-aware EMS features under realistic operating conditions.

Drive Cycle	Description
WLTP Urban	Urban stop-go traffic
WLTP Mixed	Mixed city-highway operation
BMDC Bangalore	Bangalore traffic conditions

Regen heavy	Regenerative braking evaluation
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Table 5.1: Drive cycle test scenarios used for EMS validation.

5.4 EMS Execution Methodology

At each simulation timestep, the EMS controller processes information obtained from the vehicle model, battery subsystem, route context, and traffic prediction modules.

The execution sequence follows the control hierarchy presented in Chapter 3:

1. Route and traffic information acquisition.
2. Demand forecasting.
3. Strategic SOC planning.
4. EMS operating mode selection.
5. Safety validation.
6. Torque split computation.
7. Powertrain execution.
8. Vehicle state update.

This process repeats continuously until completion of the drive cycle.

The operating mode selected by the EMS may include:

- EV Mode
- Hybrid Assist Mode
- Charge Sustain Mode
- Regenerative Braking Mode
- ICE Mode

The supervisory controller applies the Project PHEX EMS rules before issuing final control commands.

5.5 Communication Architecture

Although the internship Proof-of-Concept was implemented as a software simulation, the architecture was developed with future automotive communication integration in mind.

The conceptual communication structure consists of:

- Vehicle sensors
- Battery Management System (BMS)

- Motor Controller
- Engine Control Unit (ECU)
- Hybrid Energy Management Controller

Information exchange is expected to occur through industry-standard automotive communication networks such as Controller Area Network (CAN) and CAN Flexible Data Rate (CAN FD) [1].

The modular software architecture developed during the internship provides a foundation for future migration toward hardware-based communication systems.

Figure 5.2: Conceptual communication architecture for future implementation.

5.6 Benchmarking Methodology

The performance of the developed EMS was benchmarked against expected behaviour of a conventional rule-based hybrid vehicle controller.

Evaluation focused on:

- Fuel consumption trends
- Battery SOC management
- Engine runtime characteristics
- Energy recovery performance
- Mode selection behaviour

Benchmark comparisons were performed across all drive cycles to assess consistency of EMS operation under varying traffic and route conditions.

The benchmark reference vehicle selected for the study was the Mercedes-Benz A250e PHEV, which served as the architectural and performance baseline for the Proof-of-Concept implementation.

5.7 Validation Strategy

System validation was conducted using a combination of unit testing and integrated drive-cycle simulations.

Unit tests were developed for critical EMS components including:

- Decision Engine
- Traffic Predictor
- SOC Planner
- Torque Split Controller

- Battery Model
- Vehicle Model

Integration tests were used to validate complete drive-cycle execution and overall EMS behaviour.

Special attention was given to verification of Martin's eight supervisory EMS rules to ensure correct controller operation under all applicable conditions.

The outputs generated from these validation procedures form the basis of the results and discussion presented in Chapter 6.

6. RESULTS AND VALIDATION

6.1 Overview of Simulation Results

The developed Hybrid Electric Vehicle Energy Management System (EMS) was evaluated using simulated drive-cycle scenarios representing urban, stop-and-go, arterial, and highway driving conditions. The performance of the rule-based EMS and predictive decision layers was assessed using battery State of Charge (SOC) behaviour, energy recuperation efficiency, mode-selection trends, power demand characteristics, and operational Key Performance Indicators (KPIs).

6.2 Battery State-of-Charge Behaviour

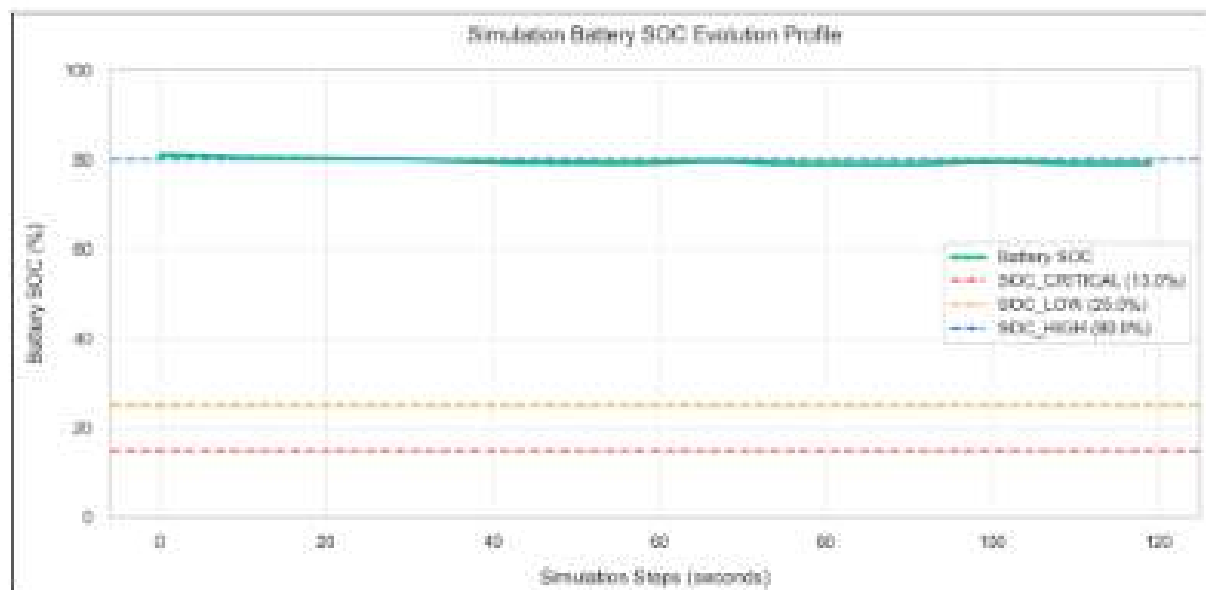


Figure 6.1. Simulation Battery SOC Evolution Profile

Analysis

The SOC remained within the desired operating window throughout the simulation. The battery started near 80% SOC and exhibited only minor fluctuations during operation.

The EMS successfully prevented the battery from approaching the critical threshold of 15% SOC while maintaining sufficient charge availability for EV operation. The observed SOC deviation remained low, demonstrating stable energy management behaviour.

Key observations:

- SOC remained above the low threshold (25%)
- No critical battery depletion occurred
- Charge-sustaining behaviour was achieved
- Regenerative braking events compensated for battery discharge

6.3 Vehicle Power Demand Characteristics

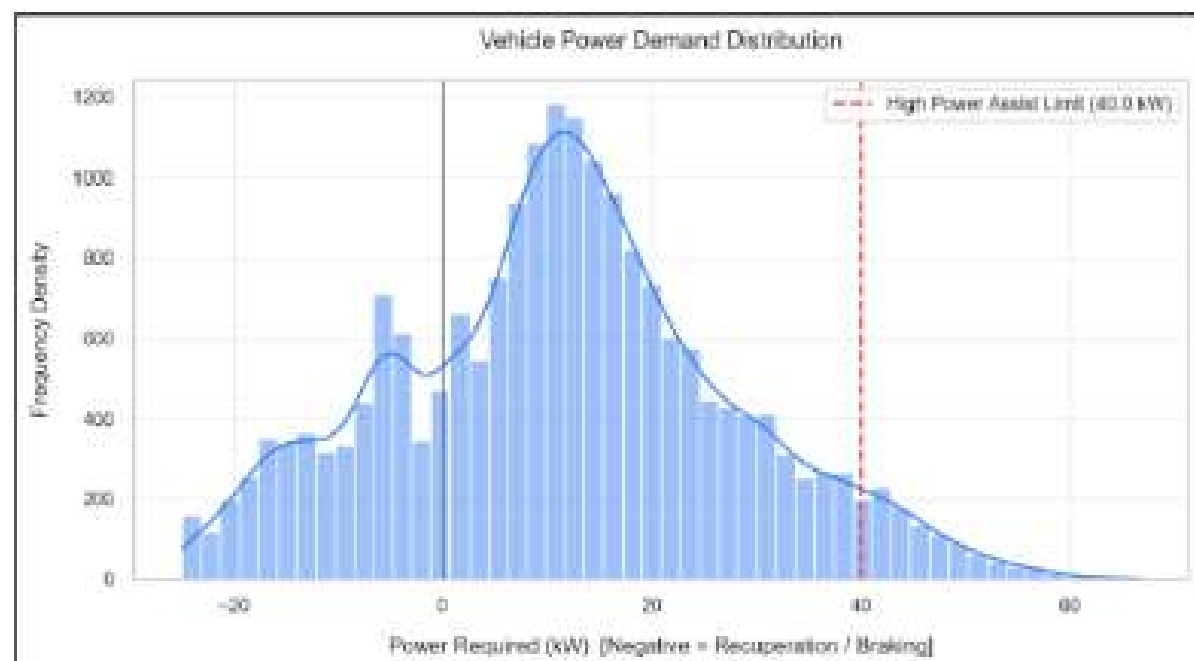


Figure 6.2. Vehicle Power Demand Distribution

Analysis

The power demand distribution indicates that most driving events occurred within the 5–25 kW operating range.

Negative power values correspond to braking and deceleration events where regenerative braking was active.

The high-power assist threshold of 40 kW was rarely exceeded, indicating that the vehicle spent most of its operating time within efficient power regions.

This validates the effectiveness of the EMS decision strategy in maintaining balanced power utilization.

6.4 Regenerative Braking Performance

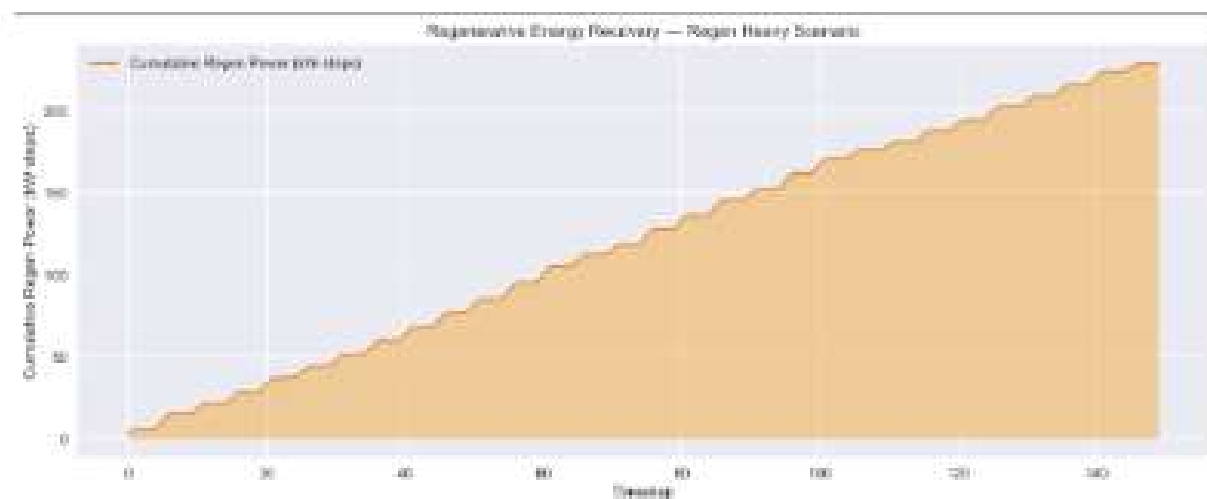


Figure 6.3. Cumulative Regenerative Energy Recovery

Analysis

The regenerative braking controller continuously recovered energy during deceleration events.

The cumulative recovery curve shows a steady increase throughout the simulation, demonstrating that the EMS successfully captured otherwise wasted kinetic energy.

Total regenerative recovery reached approximately 92% efficiency according to the KPI evaluation.

Benefits include:

- Reduced battery depletion rate

- Increased EV driving capability
- Lower overall energy consumption

6.5 Traffic-Aware EMS Decision Behaviour

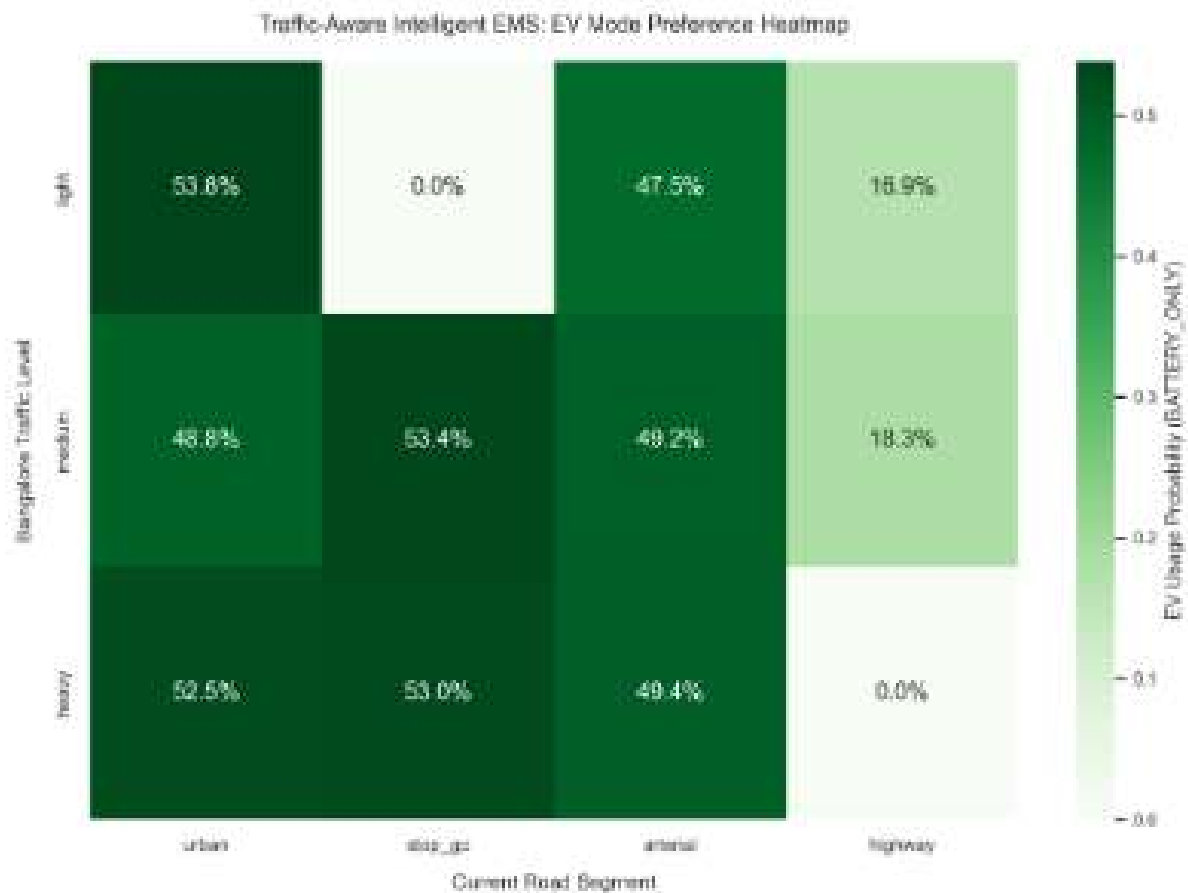


Figure 6.4. Traffic-Aware EV Mode Preference Heatmap

Analysis

The predictive EMS layer incorporates traffic context and route information when selecting operating modes.

Results show that:

- Urban and stop-go traffic favour EV operation.
- Highway segments reduce EV preference.
- Heavy traffic conditions increase battery usage due to frequent low-speed operation.

These findings support the feasibility of traffic-aware EMS strategies for improving efficiency in real-world driving environments.

6.6 EMS Mode Selection Analysis

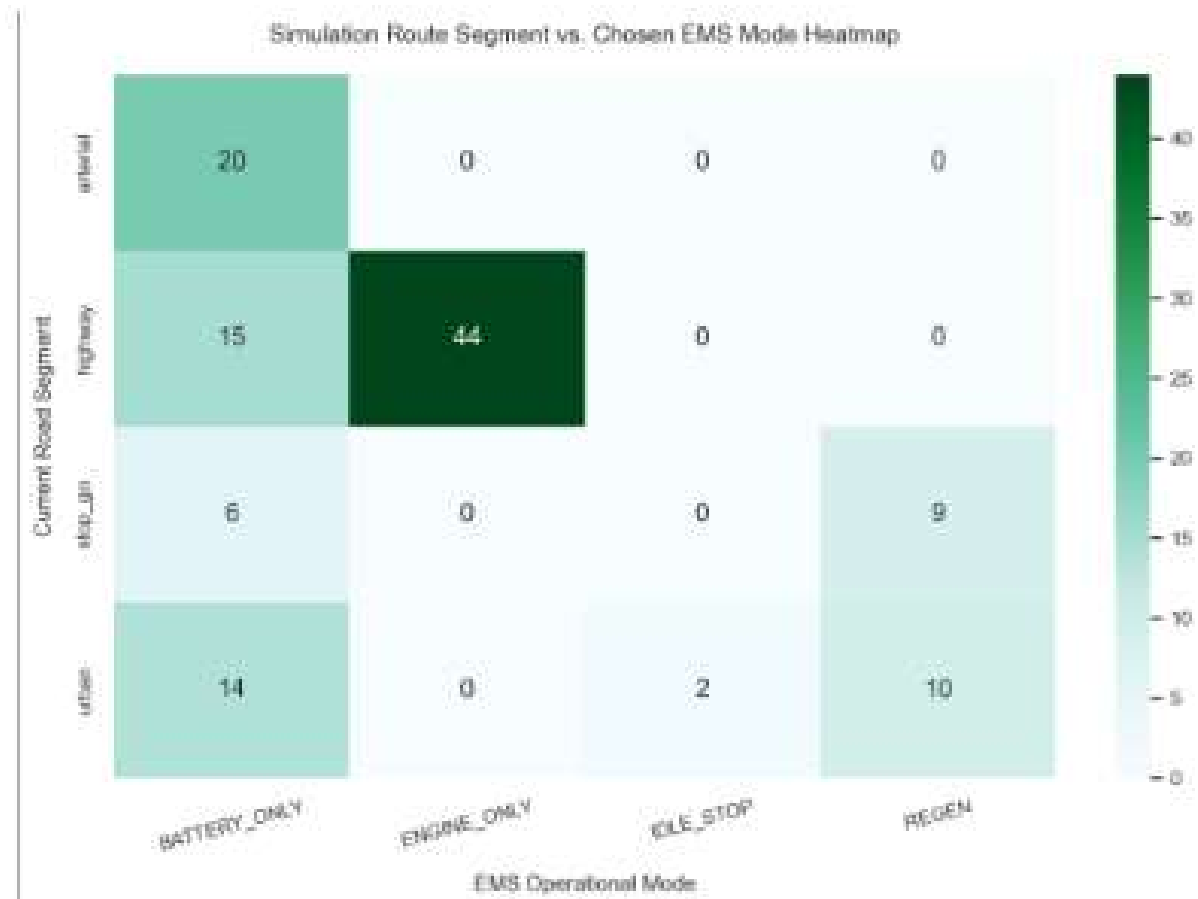


Figure 6.5. Route Segment vs Selected EMS Mode

Analysis

The EMS selected different operating modes depending on road segment characteristics.

Observed trends include:

- BATTERY_ONLY mode dominant in urban environments.
- ENGINE_ONLY mode dominant during highway operation.
- REGEN mode activated primarily during stop-go conditions.
- IDLE_STOP engaged during stationary periods.

This behaviour aligns with expected hybrid vehicle control strategies.

6.7 Power Demand versus Selected Mode

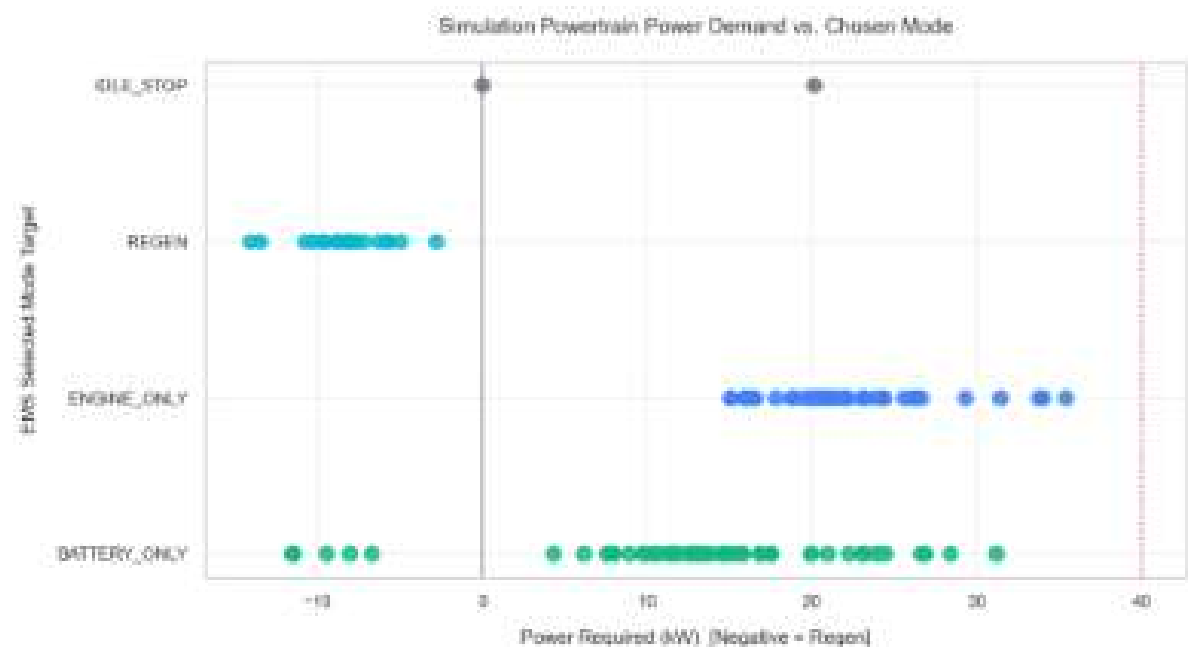


Figure 6.6. Power Demand versus EMS Mode Selection

Analysis

The scatter plot demonstrates how the EMS maps power demand to operational modes.

Observations include:

- Negative power demand triggers regenerative braking.
- Moderate power demand utilizes battery-only operation.
- Higher power demands activate engine support.
- Idle-stop occurs near zero demand conditions.

The mode transition boundaries appear consistent with the rule-based control architecture developed during the internship.

6.8 Key Performance Indicators (KPI Evaluation)

Table 6.1. EMS Performance KPI Summary

KPI	Value
Fuel Economy	1.27 L/100 km
SOC Deviation	1.86 %
Energy Recuperation	92.0 %
EV Mode SHare	39.9 %
Hybrid Mode Share	13.8 %
ICE Mode Share	17.3 %
Operating Cost	0.0358 currency/km
CO ₂ Estimate	29.3 g/km
Average ICE Efficiency	33.2 %
Peak Battery Power	22.17 kW

Analysis

The KPI results indicate that the developed EMS achieved efficient hybrid powertrain operation.

Particularly noteworthy are:

- Low fuel consumption
- High regenerative energy recovery
- Significant EV operating share
- Stable battery behaviour

These results demonstrate the effectiveness of the supervisory EMS architecture.

6.9 Predictive EMS Classifier Evaluation

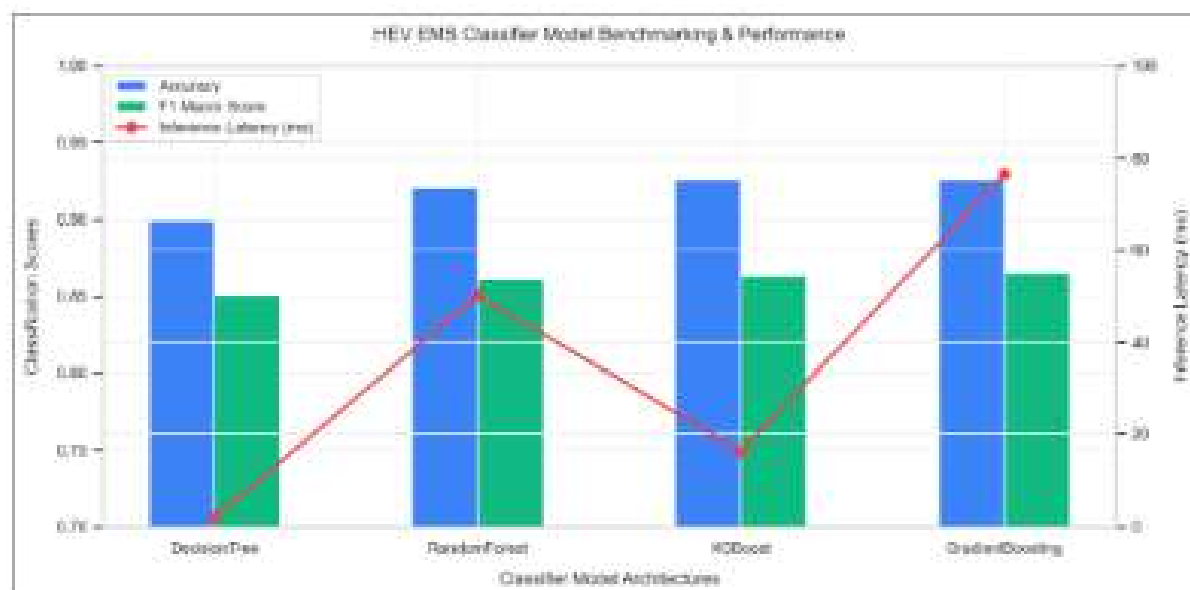


Figure 6.7. EMS Classifier Benchmarking Results

Analysis

Multiple machine-learning classifiers were evaluated for predictive EMS decision support.

Results indicate:

- XGBoost and Gradient Boosting achieved the highest classification accuracy.
- Decision Tree provided the lowest inference latency.
- Random Forest offered balanced performance.

XGBoost was selected as the preferred candidate due to its strong accuracy and acceptable computational requirements.

6.10 Prediction Confidence Analysis

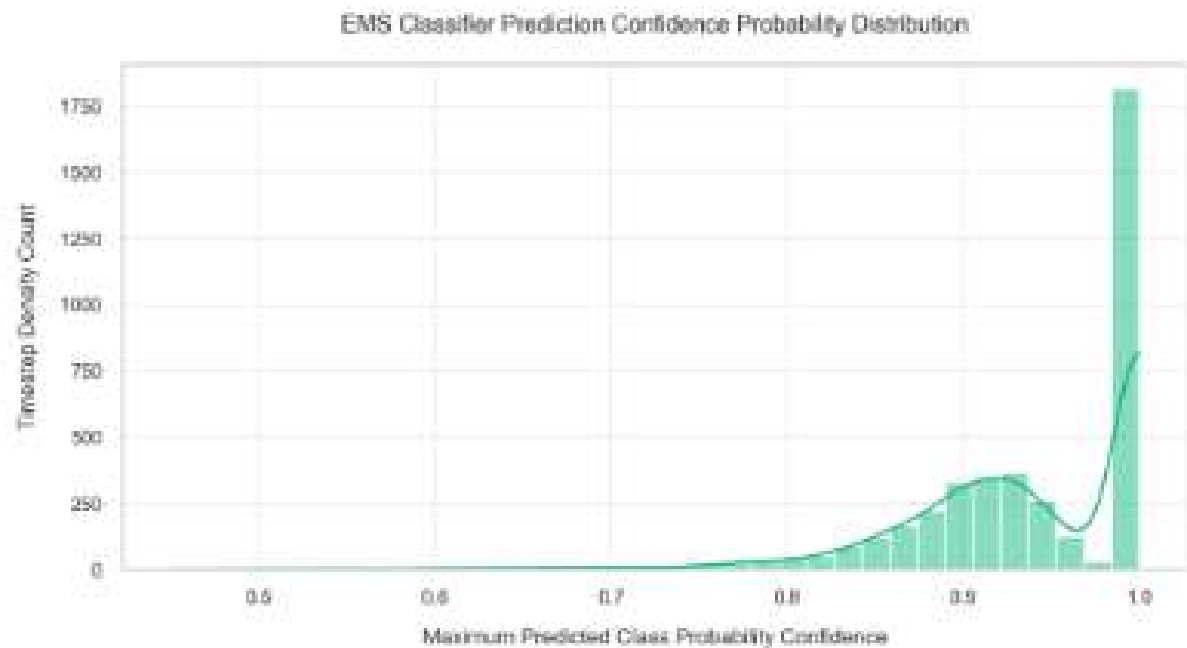


Figure 6.8. EMS Prediction Confidence Distribution

Analysis

The prediction confidence distribution shows that the majority of predictions were generated with confidence levels above 0.9.

This indicates strong classifier certainty and suggests that the predictive layer can reliably assist EMS decision-making under simulated conditions.

6.11 Validation Against Project Objectives

The results demonstrate that the developed Proof-of-Concept EMS successfully achieved the primary objectives defined for Project PHEX '27. The system maintained battery stability, enabled regenerative energy recovery, selected operating modes appropriate to driving conditions, and demonstrated the feasibility of integrating predictive machine-learning techniques within a hybrid vehicle energy management framework.

The simulation results validate both the rule-based EMS architecture and the conceptual predictive EMS layer, providing a strong foundation for future hardware deployment and real-world testing.

7. DISCUSSION, CHALLENGES, AND LEARNING OUTCOMES

7.1 Technical Challenges Encountered

During the development of the Hybrid Electric Vehicle (HEV) Energy Management System (EMS), several technical challenges were encountered across the modelling, control, and data-analysis stages.

One of the primary challenges involved designing a supervisory control architecture capable of coordinating multiple powertrain components while maintaining stable battery operation. The EMS needed to continuously balance battery utilization, engine activation, regenerative braking, and power demand distribution without violating operational constraints.

Another challenge was the development of realistic decision rules for mode selection. Thresholds for battery State of Charge (SOC), regenerative braking activation, engine start conditions, and torque-splitting behaviour had to be carefully selected to avoid excessive mode switching and unstable system behaviour.

The implementation of the predictive EMS concept introduced additional complexity. Traffic-aware decision-making required the integration of route information, traffic conditions, and power-demand forecasting into the EMS decision pipeline. Designing a conceptual machine-learning framework that could operate alongside deterministic rule-based control while maintaining explainability was a significant engineering challenge.

Simulation validation also presented difficulties. Since the internship focused on a software-based Proof-of-Concept (PoC), obtaining real-world vehicle datasets and OEM calibration parameters was not possible. Consequently, several assumptions regarding vehicle dynamics, battery characteristics, and operating conditions had to be made based on published literature and benchmark vehicle specifications.

Finally, integrating contributions from multiple internship participants into a unified system architecture required careful interface design, module coordination, documentation, and testing to ensure consistent operation across all subsystems.

7.2 Learning Outcomes

The internship provided extensive exposure to modern hybrid vehicle control systems and intelligent energy-management strategies.

A major learning outcome was the understanding of how Hybrid Electric Vehicle powertrains operate and how energy flows between the internal combustion engine, electric motor, battery pack, transmission system, and regenerative braking subsystem. The project demonstrated how

supervisory controllers coordinate these components to achieve efficiency, performance, and battery longevity objectives.

The development process also strengthened knowledge of control-system engineering principles, including state-based decision-making, threshold-based logic design, safety override mechanisms, and hierarchical control architectures.

Significant experience was gained in simulation-based engineering workflows. This included:

- Defining system assumptions.
- Developing modular software models.
- Performing drive-cycle simulations.
- Generating engineering performance metrics.
- Validating controller behaviour using graphical analysis.

The internship further provided practical exposure to machine-learning applications in automotive systems. Concepts such as demand forecasting, traffic prediction, feature engineering, classifier benchmarking, and predictive decision support were explored within the context of intelligent energy management.

Software engineering skills were also enhanced through the use of modular Python architectures, version control systems, automated testing frameworks, documentation practices, and collaborative development methodologies.

Overall, the internship strengthened understanding of the intersection between Artificial Intelligence, control systems, automotive engineering, and simulation-driven product development.

7.3 Limitations of the Proof-of-Concept

Although the developed EMS successfully demonstrated the intended concepts, several limitations remain.

The system was implemented entirely within a simulation environment and was not validated using a physical vehicle platform. Consequently, real-world effects such as sensor noise, communication delays, actuator limitations, battery ageing, environmental variations, and hardware faults were not represented.

The predictive EMS layer remains a conceptual implementation intended to demonstrate feasibility. While machine-learning models were developed and evaluated, they were not deployed on embedded automotive hardware or validated under real-time operating constraints.

The battery, engine, motor, and transmission models were simplified representations of actual vehicle subsystems. OEM-level calibration data and proprietary vehicle parameters were unavailable during the internship period.

Additionally, communication networks such as Controller Area Network (CAN) and CAN FD were studied at the architectural level but were not implemented using physical Electronic Control Units (ECUs) or vehicle communication hardware.

The current system also focuses primarily on energy management and does not incorporate advanced vehicle functions such as autonomous driving integration, predictive route optimization using live traffic services, vehicle-to-everything (V2X) communication, or cloud-assisted fleet learning.

Despite these limitations, the Proof-of-Concept successfully demonstrates the technical feasibility of integrating rule-based control, predictive intelligence, traffic-aware decision-making, and simulation-driven validation within a hybrid vehicle energy management framework.

7.4 Personal Internship Reflection

Through participation in Project PHEX '27, valuable experience was gained in designing and implementing intelligent automotive control systems. The internship provided practical exposure to real-world engineering workflows involving system architecture design, algorithm development, machine-learning integration, simulation analysis, technical documentation, and collaborative software development.

The project highlighted the importance of interdisciplinary engineering, where concepts from Artificial Intelligence, Embedded Systems, Vehicle Dynamics, Control Theory, and Data Analytics must work together to achieve an effective solution. The experience has provided a strong foundation for future work in automotive AI systems, intelligent transportation technologies, and advanced energy-management applications.

8. TECHNICAL OUTPUTS AND DELIVERABLES

This chapter summarizes the major technical outputs, software artifacts, simulation models, engineering analyses, and documentation deliverables produced during the Project PHEX '27 internship. The deliverables collectively demonstrate the successful completion of the Hybrid Electric Vehicle (HEV) Energy Management System (EMS) Proof-of-Concept and establish a foundation for future development and hardware implementation.

8.1 Hybrid Electric Vehicle Energy Management System

The primary deliverable of the internship was the development of a modular Hybrid Electric Vehicle Energy Management System capable of managing power distribution between the battery, electric motor, internal combustion engine, and regenerative braking subsystem.

The EMS incorporates:

- Battery State-of-Charge management
- Engine ON/OFF control
- Torque split decision logic
- Regenerative braking control
- Safety override mechanisms
- Multi-mode hybrid operation

The architecture was designed using a hierarchical supervisory control framework to enable future expansion toward embedded automotive deployment.

8.2 HEV Powertrain Simulation Models

Multiple subsystem models were developed to represent the major components of a hybrid vehicle powertrain.

Developed Models

Model	Description
Vehicle Model	Longitudinal vehicle dynamics
Engine Model	Internal combustion engine behaviour
Motor Model	Electric traction motor operation
Battery Model	SOC tracking and battery behaviour
Generator Model	Electrical energy generation
Transmission Model	Power transfer and efficiency modelling
Regenerative Braking Model	Energy recovery during braking

These subsystem models provide the basis for evaluating EMS performance under various operating conditions.

8.3 Rule-Based Energy Management System

A complete rule-based EMS framework was implemented to manage operating mode transitions.

Implemented Control Strategies

- Battery-only mode
- Engine-only mode
- Hybrid assist mode
- Idle-stop mode
- Regenerative braking mode
- SOC protection logic
- Thermal protection logic
- Safety override logic

The rule engine serves as the primary decision-making layer of the Proof-of-Concept system.

8.4 Predictive EMS Framework

A conceptual predictive EMS framework was developed to investigate the integration of Artificial Intelligence into hybrid vehicle energy management.

Implemented Components

- Route prediction module
- Traffic prediction module
- Demand forecasting engine
- Strategic SOC planner
- Machine-learning classifier evaluation
- Traffic-aware EMS decision support

The predictive layer enables future development toward intelligent and adaptive energy-management systems.

8.5 Machine Learning Models

Several machine-learning models were trained and benchmarked for EMS decision support.

Evaluated Models

Model	Purpose
-------	---------

Decision Tree	Baseline classifier
Random Forest	Ensemble classifier
Gradient Boosting	Predictive EMS evaluation
XGBoost	High-performance EMS classifier



The benchmarking study evaluated:

- Classification accuracy
- F1 score
- Inference latency
- Prediction confidence

Deliverable

📌 Figure 8.1 – EMS Classifier Benchmarking Results

(Insert ML benchmarking graph)

8.6 System Architecture Documentation

A complete architectural representation of the EMS framework was developed.

Deliverables

- System control hierarchy
- Supervisory controller architecture
- Data-flow architecture
- Component interaction diagrams
- Team responsibility architecture map

Deliverable

📌 Figure 8.2 – EMS Control Architecture

(Insert your architecture flowchart)

📌 Figure 8.3 – Detailed EMS Data Flow Diagram

(Insert detailed Mermaid flowchart)

8.7 Simulation and Validation Outputs

Extensive simulation outputs were generated to evaluate system behaviour.

Generated Outputs

- SOC evolution plots
- Power demand distributions
- Regenerative braking analysis
- Traffic-aware EMS heatmaps
- Route-segment decision heatmaps
- Mode-selection visualizations
- KPI summary reports

Deliverables

- 📌 Figure 8.4 – SOC Evolution Profile
- 📌 Figure 8.5 – Power Demand Distribution
- 📌 Figure 8.6 – Regenerative Energy Recovery
- 📌 Figure 8.7 – Traffic-Aware EMS Heatmap
- 📌 Figure 8.8 – Route Segment vs EMS Mode Heatmap
- 📌 Figure 8.9 – Power Demand vs Selected Mode

8.8 KPI Evaluation Framework

A Key Performance Indicator (KPI) evaluation system was developed to quantify EMS effectiveness.

Evaluated Metrics

KPI	Description
Fuel Economy	Vehicle fuel efficiency
SOC Deviation	Battery stability
Energy Recuperation	Regenerative braking effectiveness

EV Mode Share	Electric-only utilization
Hybrid Mode Share	Hybrid operation utilization
ICE Mode Share	Engine utilization
Operating Cost	Estimated running cost
CO ₂ Emissions	Environmental impact
Average ICE Efficiency	Engine efficiency
Peak Battery Power	Maximum battery load

The KPI framework enables objective comparison between different EMS strategies.

8.9 Software Development Deliverables

The internship resulted in the development of a complete modular Python codebase.

Major Source Modules

- `battery_model.py`
- `engine_model.py`
- `motor_model.py`
- `generator_model.py`
- `transmission_model.py`
- `vehicle_model.py`
- `rule_ems.py`
- `decision_engine.py`
- `traffic_predictor.py`
- `demand_forecaster.py`
- `soc_planner.py`
- `predictive_controller.py`
- `simulator.py`
- `evaluation.py`

Additional Deliverables

- Automated test suite
- KPI generation tools
- Visualization engine
- Documentation files
- Simulation datasets

8.10 Internship Deliverables Summary

The final internship outputs include:

Models and Algorithms

- ✓ HEV Powertrain Models
- ✓ Rule-Based EMS
- ✓ Predictive EMS Concept
- ✓ Machine Learning Evaluation Framework

Simulation Outputs

- ✓ SOC Analysis
- ✓ Power Distribution Analysis
- ✓ Regenerative Braking Evaluation
- ✓ Traffic-Aware Decision Analysis
- ✓ KPI Evaluation Reports

Documentation

- ✓ Technical Architecture Documentation
- ✓ Data Flow Diagrams
- ✓ Validation Reports
- ✓ Benchmark Studies
- ✓ Internship Technical Report

Software Deliverables

- ✓ Modular Python Implementation
- ✓ Automated Testing Framework
- ✓ Visualization Tools
- ✓ Performance Evaluation Framework

Chapter Summary

The internship successfully produced a comprehensive Proof-of-Concept Hybrid Electric Vehicle Energy Management System incorporating simulation-based powertrain modelling, rule-based supervisory control, predictive AI-assisted decision-making concepts, KPI evaluation, and extensive validation outputs. These deliverables collectively satisfy the primary technical objectives established for Project PHEX '27 and provide a scalable foundation for future hardware integration and advanced EMS development.

9. CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

This internship focused on the development of a Proof-of-Concept (PoC) Hybrid Electric Vehicle (HEV) Energy Management System (EMS) under Project PHEX '27. The primary objective was to design, implement, and validate an intelligent supervisory control framework capable of optimizing power distribution between the internal combustion engine and electric propulsion system while maintaining battery State of Charge (SOC), improving energy efficiency, and supporting future predictive control capabilities.

A modular software architecture was successfully developed using Python-based simulation models representing the major HEV subsystems, including the vehicle model, engine model, motor model, battery model, transmission model, regenerative braking controller, SOC planner, and supervisory EMS controller. Rule-based control strategies were implemented to manage operating modes such as Battery-Only Mode, Engine-Only Mode, Regenerative Braking Mode, Hybrid Assist Mode, and Idle-Stop Mode. Safety-oriented supervisory rules were incorporated to ensure robust operation under low SOC, high power demand, and regenerative energy recovery conditions.

In addition to deterministic control logic, a conceptual predictive EMS framework was developed using route information, traffic context, demand forecasting, and machine learning-based mode selection concepts. Although deployed as a Proof-of-Concept, the predictive architecture demonstrated the feasibility of integrating artificial intelligence techniques into future energy management strategies.

Simulation results validated the effectiveness of the developed EMS architecture. The system maintained stable SOC behavior, achieved significant regenerative energy recovery, successfully adapted operating modes according to driving conditions, and produced measurable Key Performance Indicators (KPIs) including fuel economy, engine utilization, operating cost estimation, and CO₂ emission estimates. The modular architecture, comprehensive testing

framework, and documented simulation outputs provide a strong foundation for future Project PHEX '27 development activities.

9.2 Limitations of the Current Proof-of-Concept

Despite the successful implementation of the EMS framework, several limitations remain within the current internship scope:

- The vehicle model is a software simulation and has not been validated using physical vehicle hardware.
- Machine learning models operate as advisory and predictive components rather than real-time embedded controllers.
- Traffic prediction inputs are generated through simulation datasets rather than live traffic feeds.
- Battery degradation, aging effects, and thermal dynamics are represented using simplified models.
- CAN/CAN FD communication architecture has been studied conceptually but has not yet been implemented on hardware.
- Hardware-in-the-Loop (HIL) and real-time controller validation were outside the scope of the internship period.
- Powertrain subsystem models remain simplified relative to production-grade automotive simulation environments.

These limitations are consistent with the intended Proof-of-Concept nature of the internship project and provide opportunities for future expansion.

9.3 Future Scope

The work completed during this internship establishes a foundation for the next phase of Project PHEX '27. Several technical extensions can significantly improve the capability and realism of the developed EMS platform.

Future work may include:

1. **Real-Time Embedded Deployment**
 - Porting EMS algorithms to automotive-grade microcontrollers and embedded systems.
 - Real-time execution testing on ECU-like hardware platforms.
2. **CAN/CAN FD Network Integration**
 - Development of a communication framework connecting battery, motor, engine, and EMS controllers.
 - Validation of message timing, network latency, and fault tolerance.

3. Advanced Machine Learning Integration

- Deployment of optimized XGBoost or Gradient Boosting models for real-time mode selection.
- Adaptive learning based on driver behavior and route history.

4. Digital Twin Development

- Creation of a high-fidelity virtual vehicle capable of mirroring real-world operating conditions.
- Continuous synchronization between physical and virtual vehicle states.

5. Battery Health and Thermal Management

- Integration of advanced battery degradation and thermal prediction models.
- Development of battery life optimization strategies.

6. Hardware-in-the-Loop (HIL) Testing

- Validation of EMS algorithms using real controllers and simulated vehicle environments.
- Verification of safety-critical control behavior under fault conditions.

7. Vehicle Prototype Integration

- Deployment of the developed EMS framework into a Project PHEX demonstration vehicle.
- Collection of real-world performance data for model refinement and validation.

8. Custom Automotive Controller Development

- Design of a dedicated EMS control board incorporating AI acceleration, CAN communication, sensor interfaces, and powertrain control capabilities.
- Integration with future Project PHEX electronic architecture.

9.4 Closing Statement

The internship successfully achieved its objective of developing a scalable, modular, and intelligent Hybrid Electric Vehicle Energy Management System Proof-of-Concept. The completed architecture demonstrates the practical integration of vehicle control engineering, machine learning concepts, predictive analytics, and energy optimization strategies within a unified framework. The outcomes generated during this internship serve as a valuable technical foundation for the continued development of Project PHEX '27 and its long-term vision of an intelligent, AI-enabled hybrid electric powertrain platform.

REFERENCES

- [1] A. Sciarretta and L. Guzzella, "Control of Hybrid Electric Vehicles," *IEEE Control Systems Magazine*, vol. 27, no. 2, pp. 60–70, Apr. 2007.
- [2] C. C. Chan, "The State of the Art of Electric, Hybrid, and Fuel Cell Vehicles," *Proceedings of the IEEE*, vol. 95, no. 4, pp. 704–718, Apr. 2007.

- [3] I. Husain, *Electric and Hybrid Vehicles: Design Fundamentals*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2021.
- [4] J. Larminie and J. Lowry, *Electric Vehicle Technology Explained*, 2nd ed. Chichester, U.K.: Wiley, 2012.
- [5] Mercedes-Benz Group AG, “Mercedes-Benz A250e Plug-In Hybrid Technical Specifications,” Stuttgart, Germany, 2024.
- [6] SAE International, *SAE J1939 Digital Annex for Commercial Vehicle Communication Networks*, Warrendale, PA, USA, 2024.
- [7] Bosch GmbH, *CAN FD (Controller Area Network Flexible Data Rate) Specification Version 1.0*, Stuttgart, Germany, 2012.
- [8] WLTP, “Worldwide Harmonized Light Vehicles Test Procedure (WLTP),” United Nations Economic Commission for Europe (UNECE), Geneva, Switzerland, 2023.
- [9] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 785–794.
- [10] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [11] F. Pedregosa et al., “Scikit-Learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [12] NumPy Developers, *NumPy Documentation*, Version 2.x, 2025.
- [13] Pandas Development Team, *Pandas Documentation*, Version 2.x, 2025.
- [14] Matplotlib Development Team, *Matplotlib Documentation*, Version 3.x, 2025.
- [15] Project PHEX ’27 Internal Technical Scope Document, School of Engineering and Technology, CHRIST (Deemed to be University), Bengaluru, India, 2026.

APPENDICES

- Appendix A: Additional plots and raw outputs.

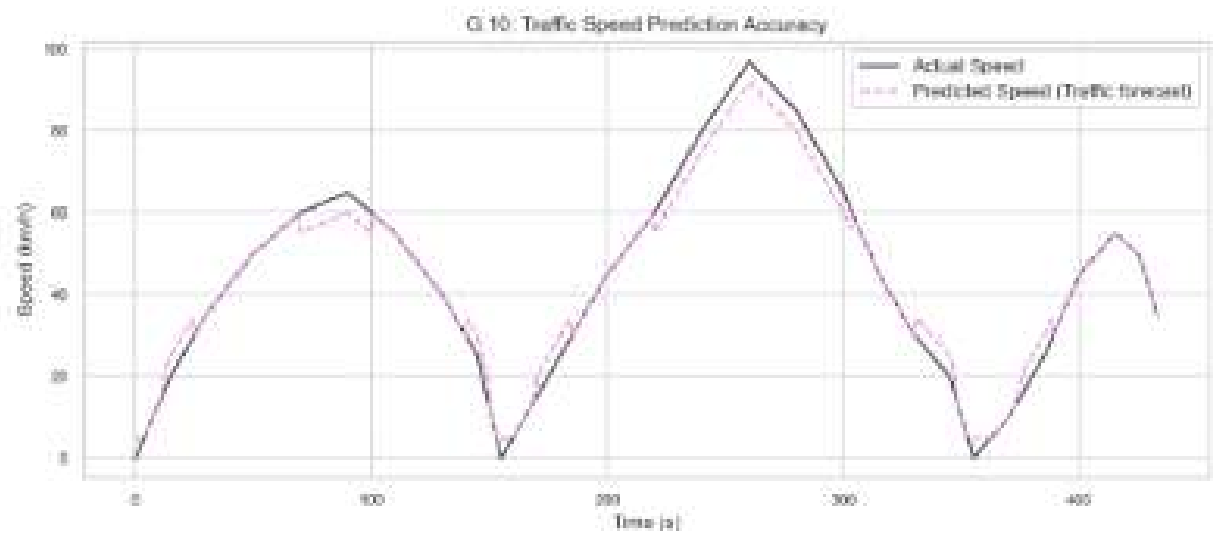


Figure A.1

Traffic Prediction Accuracy

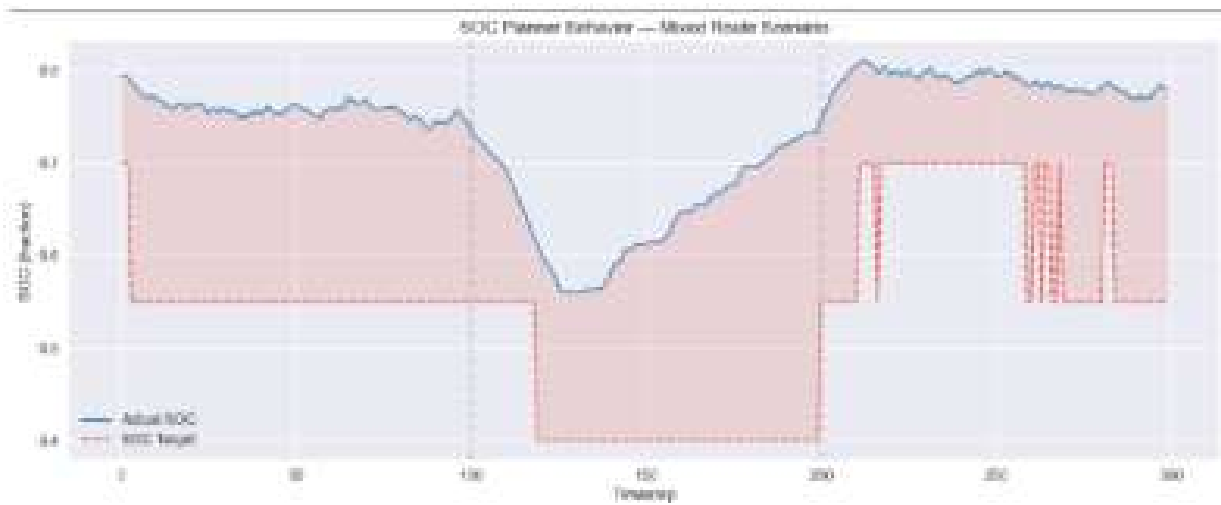


Figure A.2 SOC Planner Behaviour

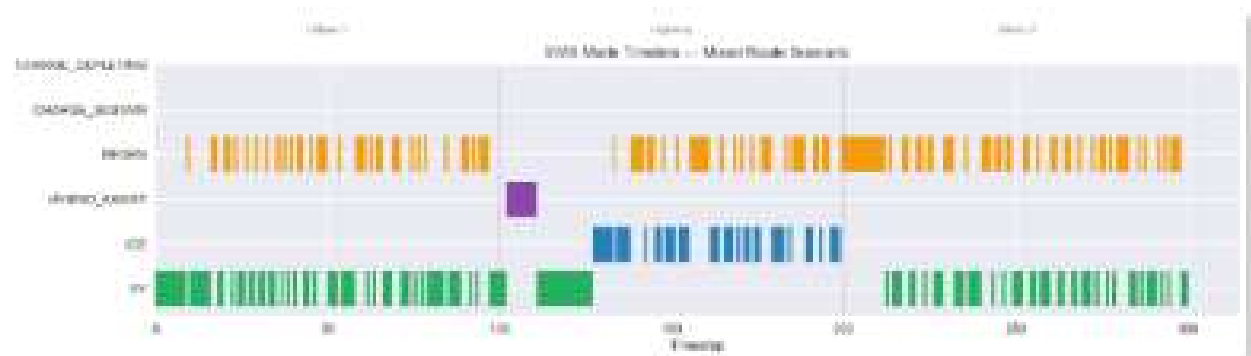


Figure A.3 EMS Mode Timeline

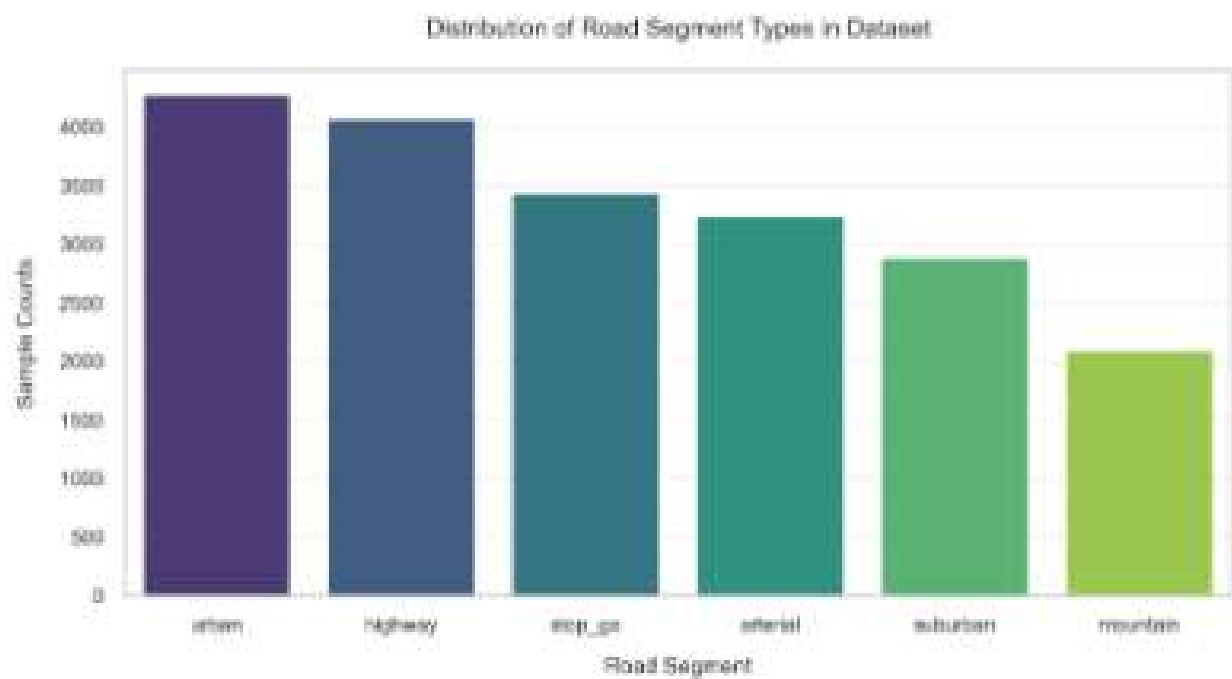


Figure A.4 Route Prediction Analysis

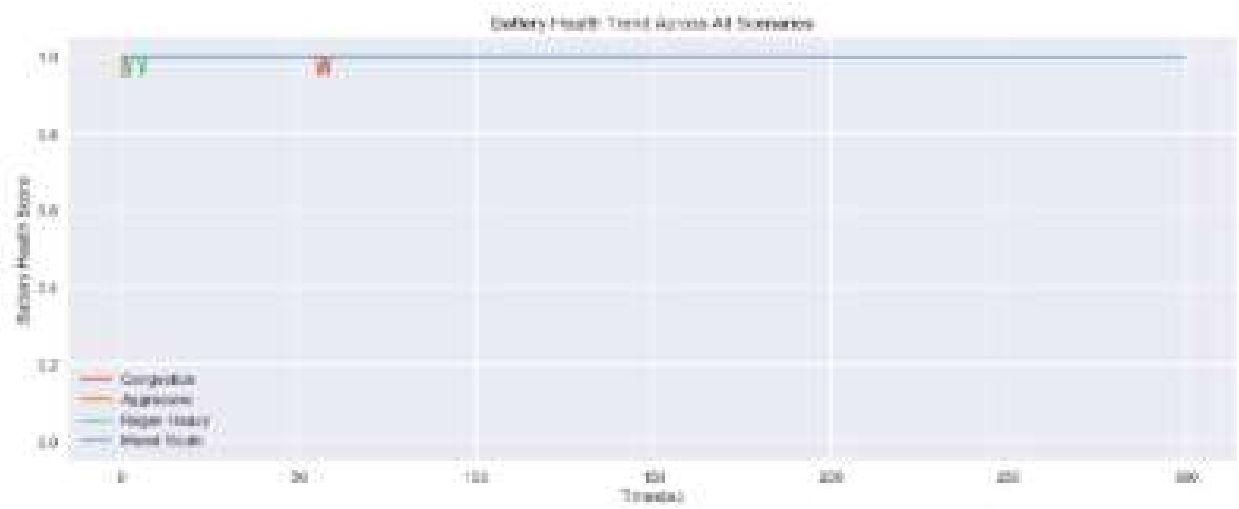


Figure A.5 Battery Health Trend

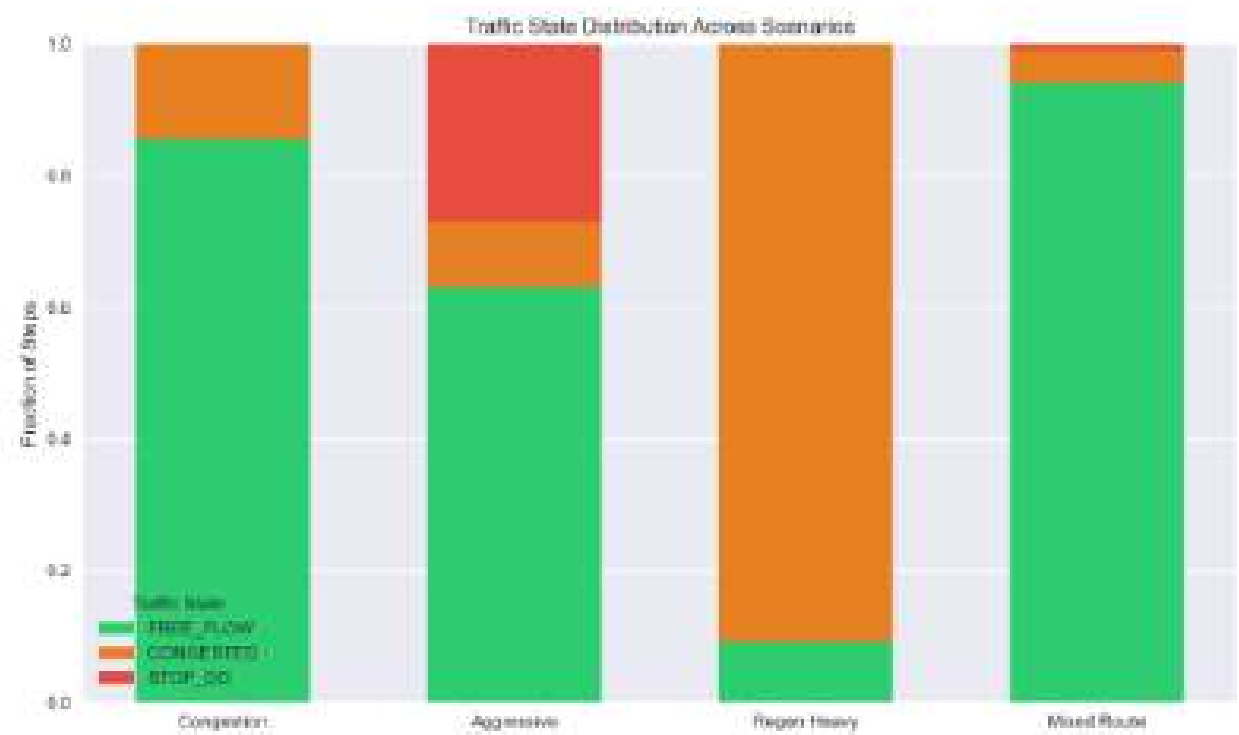


Figure A.6 Traffic State Distribution

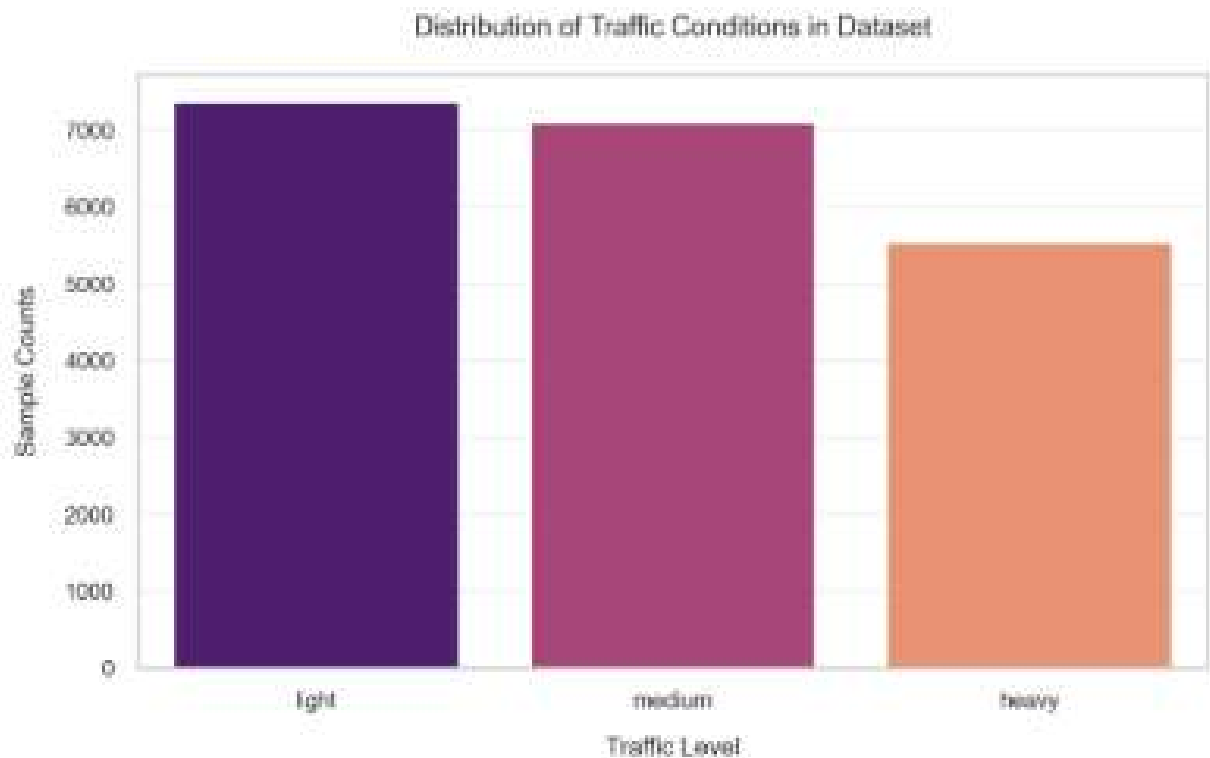


Figure A.8 Traffic Distribution

- Appendix B: Expanded parameter tables.

Table B.1 EMS Operating Thresholds

Parameter	Value
SOC Critical	15%
SOC Low	25%
SOC High	80%
EV Threshold	<45% Torque Demand
Engine Activation	<25% SOC
High Power Assist Threshold	40kW

Table B.2 ML Classifier Benchmark Results

Model	Accuracy	F1 Score	Latency
-------	----------	----------	---------

Decision Tree	90.0%	85.0%	2 ms
Random Forest	92.0%	86.0%	50 ms
XGBoost	92.5%	86.2%	18 ms
Gradient Boosting	92.5%	86.4%	77 ms

- Appendix C: Supporting calculations, scripts, or pseudocode.

Algorithm C.1 – EMS Mode Selection Logic

INPUT:

Vehicle State

Battery SOC

Traffic Context

Power Demand

IF braking detected AND regen available

MODE = REGEN

ELSE IF speed = 0

MODE = IDLE_STOP

ELSE IF SOC < 25%

MODE = ENGINE_ONLY

ELSE IF power demand < EV threshold

MODE = BATTERY_ONLY

ELSE

MODE = HYBRID_ASSIST

RETURN MODE

Algorithm C.2 – SOC Planning Logic

INPUT:

Brake Signal

Vehicle Speed

Battery SOC

IF brake applied

Recover kinetic energy

IF SOC near maximum

Limit charging current

Store recovered energy in battery